Definite integrals

1

a 33 **b** -12 **c** 46 **d** -11

e -9 **f** 192 **g** 24 **h** 84

i 15 **j** 30 **k** 490 **l** 48

m 64 **n** $-\frac{729}{4}$ **o** $\frac{38}{3}$ **p** $\frac{59}{3}$

q 1010 **r** 78 **s** $\frac{83}{24}$

Properties of definite integrals

2

a 0 **b** 0 **c** 0 **d** 0

3

a 9 **b** 45 **c** 45

d Function multiplied by a constant: $\int_a^b kf\left(x\right)dx = k\int_a^b f\left(x\right)dx$

4 10

5 13

6

a 144 **b** -144

c Switch end points: $\int_a^b f\left(x\right)dx = -\int_b^a f\left(x\right)dx$

7

a -3 **b** 9 **c** 13

8

a 12 **b** -12 **c** 32

9

a 3 **b** -5 **c** 12

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Signed areas

10

a 2 **b** 6 **c** 1 **d** 9 units^2

e $\int_0^6 f(x)\,dx$

11

a 24 **b** 10 **c** 4 **d** 38 units^2

e $\int_0^8 f(x)\,dx$

12

a -15 **b** -6 **c** -21 **d** 21 units^2

e $-\int_0^7 f(x)\,dx$

13

a 8 **b** -18π **c** $8-18\pi$ **d** $(8+18\pi) \text{ units}^2$

e $\int_0^4 f(x)\,dx - \int_4^{16} f(x)\,dx$

14

a Below **b** -35 **c** 35 units^2

15

a $x=4$ **b** $x>4$ **c** -36 **d** 36 units^2

16

a $x=3, x=9$ **b** $3 \leq x \leq 9$ **c** -36 **d** 36 units^2

17

a $x=-2, x=-8$ **b** $-8 \leq x \leq -2$ **c** 36 **d** 36 units^2

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Area under curve

18 21 units^2

19 18 units^2

20 9 units^2

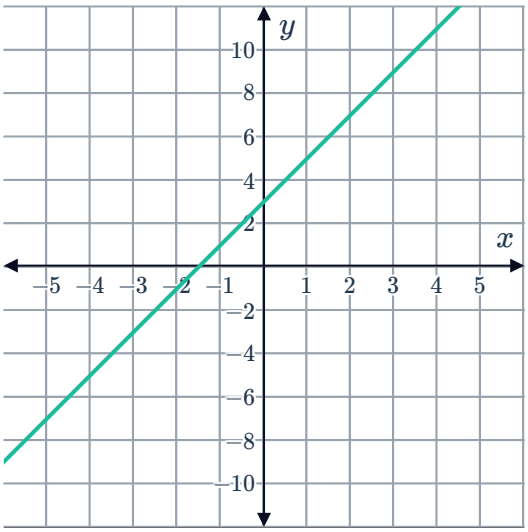
21 $\frac{23}{3}$ units^2

22 $\frac{71}{6}$ units^2

23

a

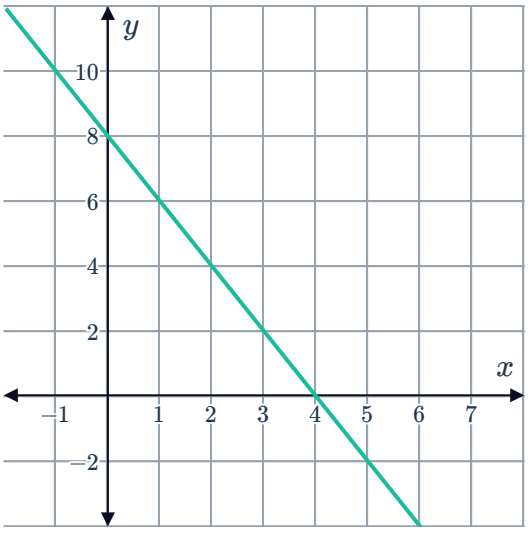
i



ii 14 units^2

b

i

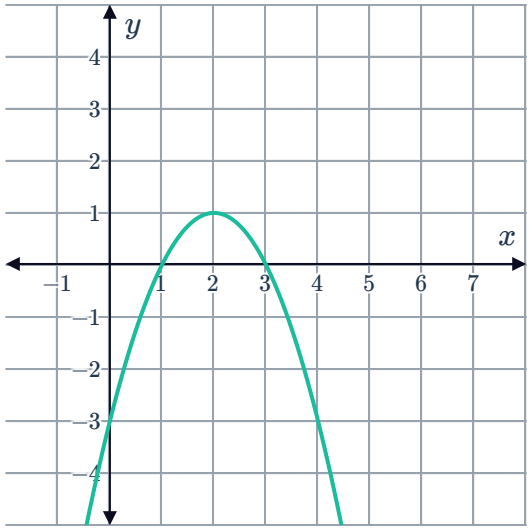


ii 8 units^2

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c

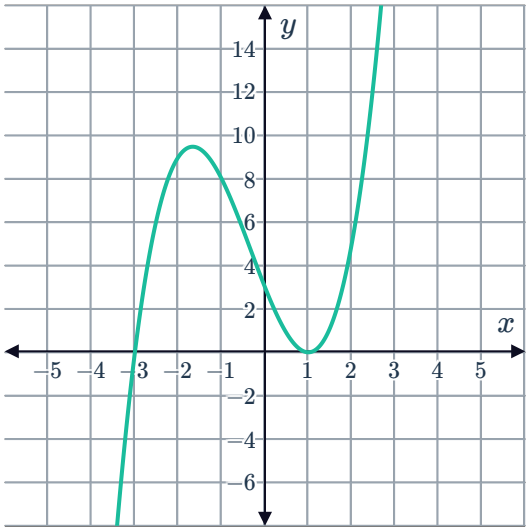
i



ii $\frac{4}{3}$ units^2

24

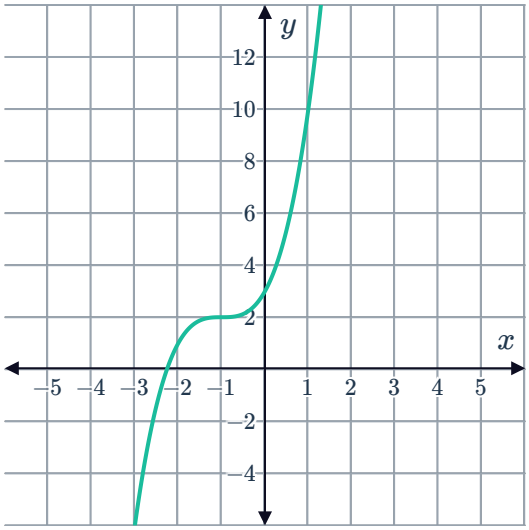
a



b $\frac{33}{4}$ units^2

25

a

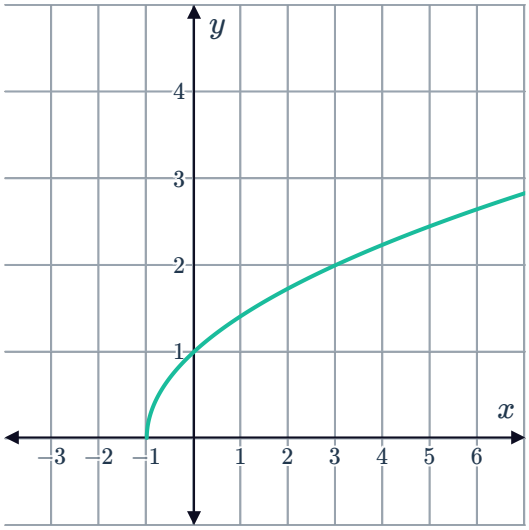


b $\frac{39}{4}$ units^2

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26

a

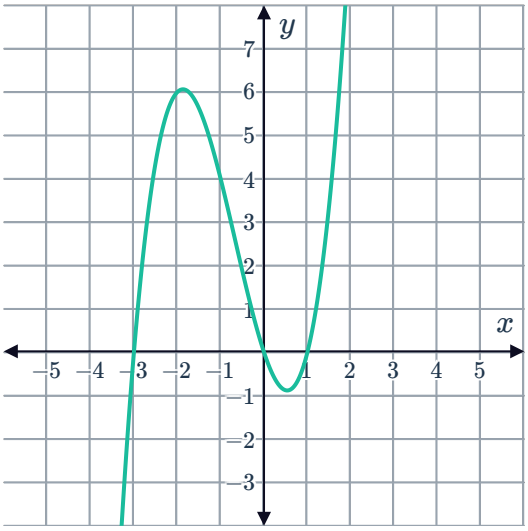


b $\frac{16}{3}$ units^2

27

a

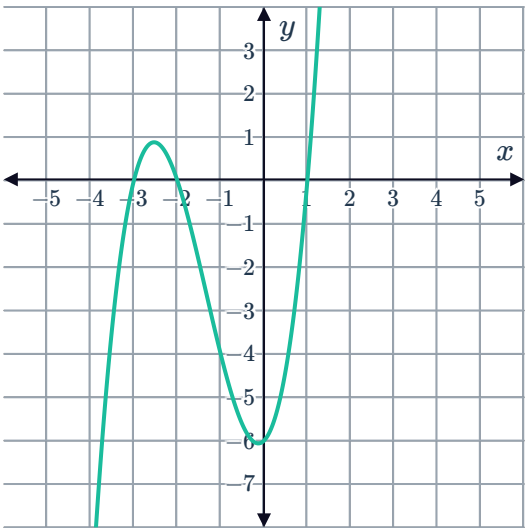
i



ii $\frac{71}{6}$ units^2

b

i

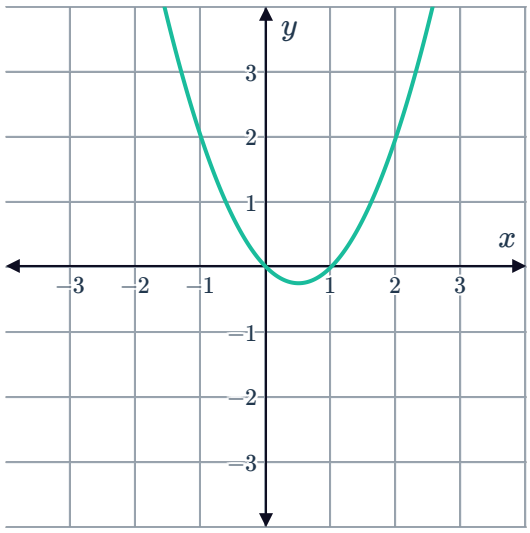


ii $\frac{71}{6}$ units^2

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c

i



ii $\frac{1}{6}$ unit^2

28

a -9 **b** -5 **c** 9 units^2 **d** 13 units^2

Area as an absolute value

29

a 8 **b** -10 **c** -2 **d** 18 units^2

e $\int_0^3 f(x)\,dx - \int_3^8 f(x)\,dx$ **f** $\int_0^8 |f(x)|\,dx$

30

a 10 **b** 14 units^2

c $\int_2^8 f(x)\,dx - \int_0^2 f(x)\,dx$ **d** $\int_0^8 |f(x)|\,dx$

31

a -1 **b** 1 **c** 13 **d** 13 units^2

32

a -2 **b** 8 **c** -4 **d** -14

e 14 **f** 14 units^2

33

a 1 **b** 6 **c** -9 **d** 10 units^2

e 10 **f** $\frac{109}{3}$

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34 **a** $-A+B$ **b** $-3A-B$ **c** $|-A+B|$ **d** $A+B$

e $-A+B-12$

35

a $B-C$ **b** $A-B+C$

c $(A+B+C)$ units^2 **d** $A+B+C$

e $35+A-B+C$

36

a -3 **b** 11 **c** 11 units^2 **d** $-\frac{163}{4}$

37

a 8 **b** 13 units^2 **c** -174

38

a $|-A+B-C|$ **b** $(|A|+|B|+|C|)$ units^2

c $-C+A$ **d** $-2A+2B-\frac{C}{2}$

e $\frac{91}{2}+A-B+C$

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